

Предел функции в точке.

Блок 1.

$$1.1. \lim_{x \rightarrow \infty} \left(2^{\frac{1}{|x|}} + \frac{\cos(\frac{1}{x})}{x^2 + 3x} \right)$$

$$1.2. \lim_{x \rightarrow \infty} \left(\ln\left(5 + \frac{2}{x}\right) + \frac{\arccos(\frac{1}{x})}{1 - 4x^2} \right)$$

$$1.3. \lim_{x \rightarrow \infty} \left(\operatorname{tg}\left(\frac{\pi}{4} - \frac{\pi}{3x^2 + 1}\right) + \frac{1 - x}{5} \right)$$

$$1.4. \lim_{x \rightarrow \infty} \left(\operatorname{ctg}\left(\frac{\pi}{x^3} + \frac{\pi}{3}\right) + \frac{\arcsin(\frac{1}{x})}{x + x^3} \right)$$

$$1.5. \lim_{x \rightarrow \infty} \left(\sin\left(\frac{\pi}{2} - \frac{\pi}{x}\right) + \frac{5}{(2x - 1)^{-1}} \right)$$

Блок 2.

$$2.1. \lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x - 3} - 1}$$

$$2.2. \lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{1 - \sqrt[4]{x}}$$

$$2.3. \lim_{x \rightarrow 16} \frac{x - 16}{\sqrt[4]{x} - 1}$$

$$2.4. \lim_{x \rightarrow 1} \frac{\sqrt[4]{x} - 1}{\sqrt[5]{x} - 1}$$

$$2.5. \lim_{x \rightarrow -1} \frac{4x^2 + 5x + 1}{x^3 + 1}$$

Блок 3.

$$3.1. \lim_{x \rightarrow \infty} \frac{7x^2 - 4x + 7}{\sqrt{x^4 + 6x^2 - 7}}$$

$$3.2. \lim_{x \rightarrow \infty} \frac{\sqrt{x^6 - x^2 - 1}}{x^4 - 9x^3 - 1}$$

$$3.3. \lim_{n \rightarrow \infty} \frac{n^6 + 3n^4 + 2}{\sqrt{3n^{10} + n^9 + 1}}$$

$$3.4. \lim_{n \rightarrow \infty} \frac{(5n^3 - 1)^2(2n^2 - 7)^3}{10\sqrt{n^{12} - 8}}$$

$$3.5. \lim_{x \rightarrow \infty} \frac{(x^4 + x^2 + 1)^{25}}{(2x + 1)^5(x^4 + 1)^5}$$

Блок 4.

$$4.1. \lim_{x \rightarrow \infty} (3x - \sqrt{9x^2 + 4x})$$

$$4.2. \lim_{x \rightarrow \infty} (2\sqrt{x^2 + 2} - 4x^2)$$

$$4.3. \lim_{n \rightarrow \infty} (\sqrt{8n^2 + 3n} - 2\sqrt{2n})$$

$$4.4. \lim_{n \rightarrow \infty} (3n - \sqrt{(3n - 2)(3n + 2)})$$

$$4.5. \lim_{n \rightarrow \infty} (\sqrt{n^2 + 3n} - \sqrt{n^2 - 4})$$

Блок 5.

$$5.1. \lim_{x \rightarrow 0} \frac{e^{x+1} - e}{x}$$

$$5.2. \lim_{x \rightarrow 1} \frac{x - 1}{2^{x-1} - 1}$$

$$5.3. \lim_{x \rightarrow 0} \frac{\ln(1 - x)}{2x}$$

$$5.4. \lim_{x \rightarrow 1} \frac{x - 1}{\log_2 x}$$

$$5.5. \lim_{x \rightarrow 1} \frac{\sin(x - 1)}{3(x - 1)}$$

Блок 6.

$$6.1. \lim_{x \rightarrow \infty} \left(\frac{x - 4}{x + 5} \right)^{\frac{2x-1}{7}}$$

$$6.2. \lim_{x \rightarrow \infty} \left(\frac{x^2 - 2}{x^2 + 3} \right)^{2x^2+6}$$

$$6.3. \lim_{n \rightarrow \infty} \left(\frac{3 - 2n}{5 - 2n} \right)^{\frac{n-6}{5}}$$

$$6.4. \lim_{n \rightarrow \infty} \left(\frac{n^2 + 3}{n^2 - 2} \right)^{n^2-3}$$

$$6.5. \lim_{n \rightarrow \infty} \left(\frac{5n + 7}{5n - 2} \right)^{\frac{3n+1}{5}}$$

Блок 7.

$$7.1. \lim_{x \rightarrow -0} \frac{\sqrt{1 - \cos 6x}}{x}$$

$$7.2. \lim_{x \rightarrow 0} \frac{x \cdot \arcsin\left(\frac{x}{2}\right)}{1 - \cos 4x}$$

$$7.3. \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{\sqrt{x^4 + 1} - 1}$$

$$7.4. \lim_{x \rightarrow 0} \frac{\sin 3x}{3 - \sqrt{2x^2 + 9}}$$

$$7.5. \lim_{x \rightarrow 0} \frac{\ln(2x + 1)}{x \cdot \operatorname{ctg}\left(x + \frac{\pi}{4}\right)}$$

Блок 8.

$$8.1. \lim_{x \rightarrow \pi} \frac{\sqrt{1 - \operatorname{tg} x} - \sqrt{1 + \operatorname{tg} x}}{\sin 2x}$$

$$8.2. \lim_{x \rightarrow \infty} x \cdot (e^{\frac{2}{x}} - 1)$$

$$8.3. \lim_{x \rightarrow 3} \frac{\ln(x^2 - 6x + 10)}{e^{(x^2-9)} - 1}$$

$$8.4. \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{\cos 2x}$$

$$8.5. \lim_{x \rightarrow -3} \frac{\ln(x^2 + 3x + 1)}{x^2 - 9}$$

Блок 9.

$$9.1. \lim_{x \rightarrow 0} \sqrt[x]{1 - 2 \sin x}$$

$$9.2. \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\frac{1}{\operatorname{ctg}^2 x}}$$

$$9.3. \lim_{x \rightarrow +\infty} (x + \sqrt{x^2 + 1})^{\frac{3}{\ln x}}$$

$$9.4. \lim_{x \rightarrow 1+0} (5x^2 - 3x - 2)^{\frac{1}{\ln(x-1)}}$$

$$9.5. \lim_{x \rightarrow 0} (e^x + \sin x)^{\frac{1}{x}}$$

$$9.6. \lim_{x \rightarrow \frac{\pi}{4}} (\operatorname{tg} x)^{\operatorname{tg} 2x}$$

$$9.7. \lim_{x \rightarrow 0} \frac{\sqrt[10]{1024 + x} - 2}{x}$$

$$9.8. \lim_{n \rightarrow 0} \frac{\ln(\sin x + \cos x)}{x}$$

$$9.9. \lim_{x \rightarrow +\infty} \frac{\ln(1 + e^{3x})}{\ln(1 + e^{7x})}$$

$$9.10. \lim_{x \rightarrow +\infty} x \cdot \operatorname{arctg} x$$